

ICTW 113-Foundations of Educational and Instructional Technologies.



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Technology Integration Planning (TIP) Model for teachers

Purpose of the TIP Model

- Designed to help teachers—especially **new or emerging technology users**
- Provides a **systematic, problem-solving approach** to technology integration
- Helps teachers select:
 - Appropriate strategies
 - Suitable digital tools
 - Effective instructional materials
- Ensures technology use is:
 - Meaningful
 - Efficient
 - Instructionally justified

Rationale for Using the TIP Model

- Technology integration must be **planned**, not incidental
- Poorly planned technology use often:
 - Adds workload without improving learning
 - Fails to solve instructional problems
- The TIP model:
 - Anchors technology decisions in **instructional needs**
 - Aligns theory, pedagogy, and practice

Overview of the TIP Model

- The TIP Model consists of **three major phases**:
- **Phase 1:** Analysis of Teaching and Learning Needs
- **Phase 2:** Design of an Integration Framework
- **Phase 3:** Post-Instruction Analysis and Revisions
- Each phase contains **specific, sequenced steps** that guide decision-making.

TIP Model – Phase 1

Phase 1: Analysis of Teaching and Learning Needs

- Focus:
- Identifying instructional problems
- Determining whether technology offers a **relative advantage**
- Steps:
- **Step 1: Determine relative advantage**
- **Step 2: Assess required resources and skills**

Step 1 – Determining Relative Advantage

- Relative advantage asks:

Is technology a better solution than current methods?

- Teachers must clearly state:
 - The instructional or learning problem
- Avoid defining problems as:
 - “Students are not using technology”
- Focus instead on:
 - Achievement gaps
 - Motivation issues
 - Conceptual difficulty
 - Inefficiency in instruction

Evidence of Instructional Problems

- Observable indicators include:
- Persistent low achievement in specific skills
- Students' lack of engagement or motivation
- Curriculum shifts toward:
 - Critical thinking
 - Problem solving
- Teacher observations and surveys
- Student resistance to traditional learning tasks

Evaluating Technology's Relative Advantage

- Teachers consider:
- Instructional benefits of technology-based strategies
- Time, effort, and cost of implementation
- Whether benefits outweigh challenges
- Alignment with:
 - Directed instruction needs
 - Constructivist learning goals

Step 2 – Assess Required Resources and Skills

- Teachers assess:
- Availability of:
 - Hardware
 - Software
 - Internet access
 - Peripherals
- Their own:
 - Technical skills
 - Classroom management skills
 - Experience with similar technology-enhanced lessons

Resource Considerations

- Key questions:
- Are there enough computers for the lesson format?
- Is the software legally available (site licenses)?
- Are printers, scanners, cameras, or probes needed?
- Can students work:
 - Individually
 - In pairs
 - In small groups?

Teacher Skill Requirements

- Teachers must be able to:
 - Demonstrate the technology effectively
 - Troubleshoot basic technical problems
 - Guide students' online behavior safely
 - Organize and manage technology-rich activities
 - Prevent lessons from becoming “technical support sessions”

TIP Model – Phase 2

Phase 2: Design of an Integration Framework

- Focus:
- Defining outcomes
- Planning instruction
- Designing assessment
- Steps:
- **Step 3: Decide on objectives and assessments**
- **Step 4: Design integration strategies**
- **Step 5: Prepare the instructional environment**

Step 3 – Decide on Objectives and Assessments

- Objectives:
- State **expected outcomes** of technology use
- Must be:
 - Observable
 - Measurable
- Can target:
 - Student achievement
 - Attitudes
 - Motivation
 - Teacher efficiency

Assessment Strategies

- Assessment methods should align with outcomes:
 - Written tests → knowledge and skill mastery
 - Rubrics → complex products (projects, multimedia)
 - Checklists → task completion
 - Surveys → attitudes and motivation
 - Observations → behaviours (e.g., engagement)

Step 4 – Design Integration Strategies

- Key design decisions:
- Directed vs. constructivist learning environment
- Single-subject or interdisciplinary approach
- Student grouping:
 - Whole class
 - Individual
 - Pairs
 - Small groups
- Preparation time for students to learn tools

Step 5 – Prepare the Instructional Environment

- Teachers must ensure:
 - Accessibility for students with disabilities (UDL principles)
 - Online safety and privacy protections
 - Availability of handouts, guides, and support materials
 - Functionality of all technology before instruction begins

TIP Model – Phase 3

Phase 3: Post-Instruction Analysis and Revisions

- Focus:
- Evaluating effectiveness
- Improving future implementation
- Steps:
- **Step 6: Analyze lesson results**
- **Step 7: Make revisions based on evidence**

Step 6 – Analyze Lesson Results

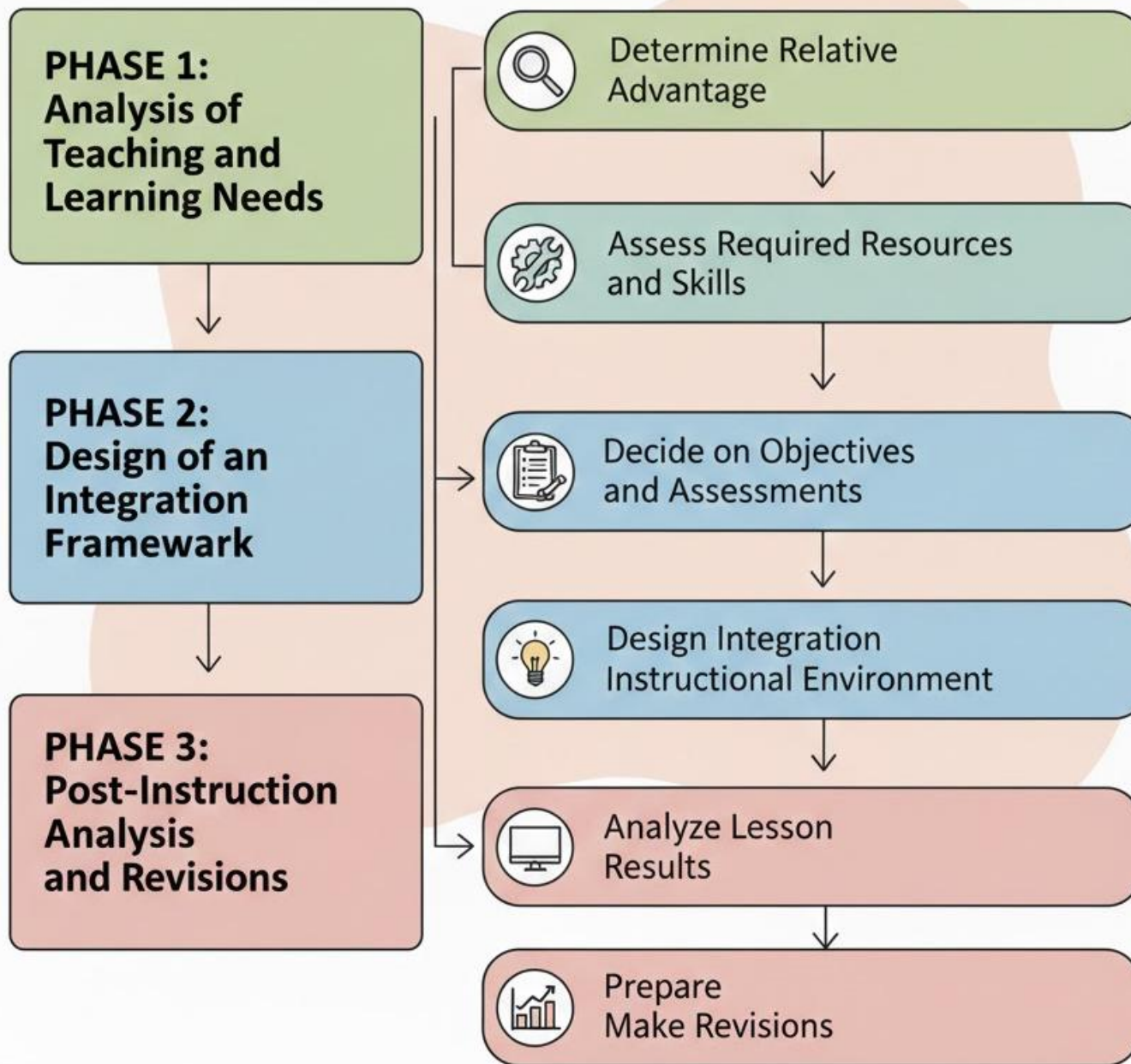
- Teachers examine:
 - Achievement data
 - Attitude and motivation data
 - Student feedback
 - Quality of student products
 - Efficiency of instruction
 - Equity of outcomes across learners

Student Feedback as Evaluation Data

- Students provide valuable insights as:
 - Participants
 - End users
- Informal discussions reveal:
 - Engagement levels
 - Frustrations
 - Suggestions for improvement
- Feedback should include:
 - High-achieving students
 - Lower-achieving students

Step 7 – Making Revisions

- Revisions may include:
 - Adjusting schedules
 - Simplifying products
 - Improving scaffolding
 - Enhancing technical skill instruction
 - Modifying assessment strategies
 - Revisiting original problem definitions if needed



Strengths of the TIP Model

- Grounded in learning theory
- Encourages reflective teaching
- Prevents technology-for-technology's-sake
- Supports novice and experienced teachers
- Promotes continuous improvement

Role of Universal Design for Learning (UDL)

- UDL supports inclusive technology integration by:
- Providing multiple means of:
 - Representation
 - Expression
 - Engagement
- Benefits students with disabilities **and all learners**
- Technology enables:
 - Text-to-speech
 - Adjustable text formats
 - Alternative assessment formats

Example of how to
implement TIP

Context

- Ama is a **Junior High School Science teacher**. She notices that her students perform poorly in **photosynthesis** questions. Many students:
 - Memorize definitions without understanding the process
 - Struggle to visualize how sunlight, carbon dioxide, and water interact
 - Lose interest during traditional chalk-and-talk lessons
- Ama decides to use **technology** to address this problem and applies the **Technology Integration Planning (TIP) Model**.

PHASE 1: Analysis of Teaching and Learning Needs

Step 1: Determine Relative Advantage

- **Problem Identified**
- Students find photosynthesis **abstract and difficult to visualize**
- Test results show consistently low scores on related questions
- Students are disengaged during lessons
- **Relative Advantage of Technology**
- An **interactive simulation** can visually demonstrate the process
- Students can manipulate variables such as light intensity and carbon dioxide levels
- Technology makes learning more engaging and conceptually clear than static diagrams
- **Decision**
 - ✓ Technology offers a **clear advantage** over traditional teaching methods.

Step 2: Assess Required Resources and Skills

- **Resources Needed**

- Computers or tablets (1 per pair of students)
- Internet access
- Photosynthesis simulation software or website
- Projector for teacher demonstration

- **Teacher Skills Required**

- Ability to demonstrate the simulation
- Classroom management skills for group work
- Basic troubleshooting skills

- **Decision**

- ✓ Required resources and skills are available and manageable.

PHASE 2: Design of an Integration Framework

Step 3: Decide on Objectives and Assessments

- **Objectives**
 - At least **80% of students** will correctly explain the process of photosynthesis
 - Students will identify **key inputs and outputs** of photosynthesis
 - Students will demonstrate improved understanding through simulation-based tasks
- **Assessment Methods**
 - Short-answer quiz
 - Worksheet based on simulation observations
 - Teacher observation checklist for engagement

Step 4: Design Integration Strategies

- **Instructional Approach**
- Combination of **directed instruction** and **constructivist learning**
 - Directed: Teacher introduces key concepts
 - Constructivist: Students explore the simulation
- **Grouping**
- Students work in **pairs** to encourage peer support
- **Learning Activities**
- Teacher demonstrates the simulation using a projector
- Students explore the simulation in pairs
- Students change variables and observe outcomes
- Groups discuss findings and answer guiding questions

Step 5: Prepare the Instructional Environment

- **Preparation Activities**

- Test the simulation before the lesson
- Prepare a worksheet with guiding questions
- Arrange seating for easy pair collaboration
- Establish Internet safety and usage rules
- Prepare a backup diagram in case of technical failure

PHASE 3: Post-Instruction Analysis and Revisions

Step 6: Analyze Lesson Results

- **Evidence Collected**
 - Quiz scores show improved performance
 - Students can verbally explain the process more confidently
 - Observation shows increased engagement
- **Student Feedback**
 - Students report that the simulation helped them “see” the process
 - Some students request more time to explore the simulation

Step 7: Make Revisions

- **Planned Improvements**
- Allocate more time for simulation exploration
- Add a short recap video before the activity
- Include a reflection question to strengthen retention
- Provide extension activities for fast learners

Key Takeaways

- Technology integration is a **problem-solving process**
- Effective integration requires:
 - Clear instructional purpose
 - Strategic planning
 - Evidence-based evaluation
- The TIP model ensures technology use is:
 - Purposeful
 - Pedagogically sound
 - Sustainable